

1. 设强度为 λ .

$$\text{则 } P(X=1) = \lambda e^{-\lambda} \quad P(X=2) = \frac{\lambda^2}{2!} e^{-\lambda}$$

$$\text{解: } \lambda = 2$$

$$\therefore P(X=4) = \frac{2^4}{4!} e^{-2}$$

5. 记 ~~记~~ 成功次数为 Y

则 Y 符合强度为 λp 的 Poisson 分布

$$\therefore P(X=Y=k) = \frac{(\lambda p)^k}{k!} e^{-\lambda p} \quad EY = \lambda p$$

6. 记 N 为虫数, X 为虫卵数

$$\therefore P(X=x) = P\left(\sum_{i=1}^N X_i = x\right)$$

$$= \sum_{k=0}^{\infty} P\left(\sum_{i=1}^N X_i = x \mid N=k\right) P(N=k)$$

$$= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} P\left(\sum_{i=1}^k X_i = x\right)$$

$$= \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} \frac{(k\mu)^x}{x!} e^{-k\mu} \quad \left(\sum_{i=1}^k X_i \text{ 符合 } k\mu \text{ 的泊松分布}\right)$$

$$= \frac{\mu^x}{x!} e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^{-\mu})^k}{k!} \lambda^x$$

$$7. P(X=k) = P(X=k|A=A_1)P(A=A_1) + P(X=k|A=A_2)P(A=A_2) \\ = p \frac{\lambda_1^k e^{-\lambda_1}}{k!} + (1-p) \frac{\lambda_2^k e^{-\lambda_2}}{k!}$$

10. (1) 独立: $X \sim \text{Poisson}_{\lambda p}$ $Y \sim \text{Poisson}_{\lambda(1-p)}$

$$(2) P(X=x, Y=y) = \sum_{k=0}^{\infty} P(X=x, Y=y|N=k)P(N=k)$$

$$= P(X=x, Y=y|N=x+y)P(N=x+y)$$

$$= \binom{x+y}{x} p^x (1-p)^y \cdot \frac{\lambda^{x+y} e^{-\lambda}}{(x+y)!} = \frac{1}{x!y!} p^x (1-p)^y \lambda^{x+y} e^{-\lambda}$$

$$P(X=x)P(Y=y) = \left(\frac{(\lambda p)^x e^{-\lambda p}}{x!} \right) \left(\frac{(\lambda(1-p))^y e^{-\lambda(1-p)}}{y!} \right) = \frac{1}{x!y!} p^x (1-p)^y \lambda^{x+y} e^{-\lambda}$$

$$\therefore P(X=x, Y=y) = P(X=x)P(Y=y) \quad \therefore X \perp Y$$

$$12. P(X_1=x, N=n) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} \sum_{i=0}^n \frac{\lambda^{n-i}}{(n-i)!} & n \geq x \\ \frac{\lambda^x}{x!} e^{-\lambda} \cdot \frac{\lambda^{n-x}}{(n-x)!} & n < x \end{cases}$$

$$\therefore P(N=n) = \sum_{i=0}^{\infty} P(X_1=i, N=n) = 2e^{-2\lambda} \frac{\lambda^n}{n!} \sum_{j=n+1}^{\infty} \frac{\lambda^j}{j!} + \frac{\lambda^{2n} e^{-2\lambda}}{(n!)^2}$$

$$P(X_1=x, N=m) = \begin{cases} \frac{\lambda^x}{x!} e^{-\lambda} \sum_{i=0}^m \frac{\lambda^{m-i}}{(m-i)!} & m \geq x \\ \frac{\lambda^x}{x!} e^{-\lambda} \cdot \frac{\lambda^{m-x}}{(m-x)!} & m < x \end{cases}$$

$$\therefore P(N=m) = 2e^{-2\lambda} \frac{\lambda^m}{m!} \sum_{j=0}^{m-1} \frac{\lambda^j}{j!} + \frac{\lambda^{2m} e^{-2\lambda}}{(m!)^2}$$

13. 不妨设 $s \leq t$

$$\begin{aligned} E(N_s, N_t) &= E[N_s((N_t - N_s) + N_s)] \\ &= E(N_s)E(N_t - N_s) + E(N_s^2) \\ &= \lambda^2 s(\lambda t - \lambda s) + \lambda s + \lambda^2 s \\ &= \lambda^2 s t + \lambda s \end{aligned}$$

$$\begin{aligned} \therefore \text{Cov}(N_s, N_t) &= E(N_s N_t) - E N_s E N_t = \lambda s \\ &= \lambda \min\{s, t\} \end{aligned}$$

15. ~~$E X =$~~

记营业量为 Y , $X \sim \text{Poisson}_{10800}$

$$E(Y) = \sum_{k=0}^{\infty} E\left(\sum_{i=1}^X \eta_i \mid X=k\right) P(X=k)$$

$$= \sum_{k=1}^{\infty} k \times 10 \times 0.15 \times \frac{(10800)^k}{k!} e^{-10800}$$

$$= 10800 \times 1.5 = 16200$$

$$E(Y^2) = \sum_{k=0}^{\infty} (k^2 + k) E\eta_1^2 P(X=k)$$

$$\therefore DY = EY^2 - (EY)^2 = \lambda t E\eta_1^2 = 38070$$

$$\therefore \sigma_Y = \sqrt{DY} = 195.12$$

$$1. (a) \int_{-1}^1 \frac{C}{\sqrt{1-x^2}} dx = 1$$

$$\text{hence } C = \frac{1}{\pi}$$

$$(b) \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\pi \sqrt{1-x^2}} dx = \frac{1}{3}$$

$$4. P(X > C) = \int_C^{+\infty} \frac{1}{\theta} e^{-\frac{x}{\theta}} dx = e^{-\frac{C}{\theta}} = \frac{1}{2}$$

$$\therefore C = \theta \ln 2$$